Resolve data import errors

Introduction

So far, you've revised how to resolve inconsistencies and data quality issues in Microsoft Power BI. In this activity, you'll apply your knowledge of the Power Query Editor to profile and resolve issues before importing data.

Case study

At Adventure Works, the fulfillment team wants to conduct an analysis of recent orders. The team manager has provided you with the dataset, and you've been tasked with importing it into Power BI. Unfortunately, the team tracks orders manually in an Excel sheet, which has resulted in human error. You need to evaluate the dataset and resolve errors before importing it into Power BI.

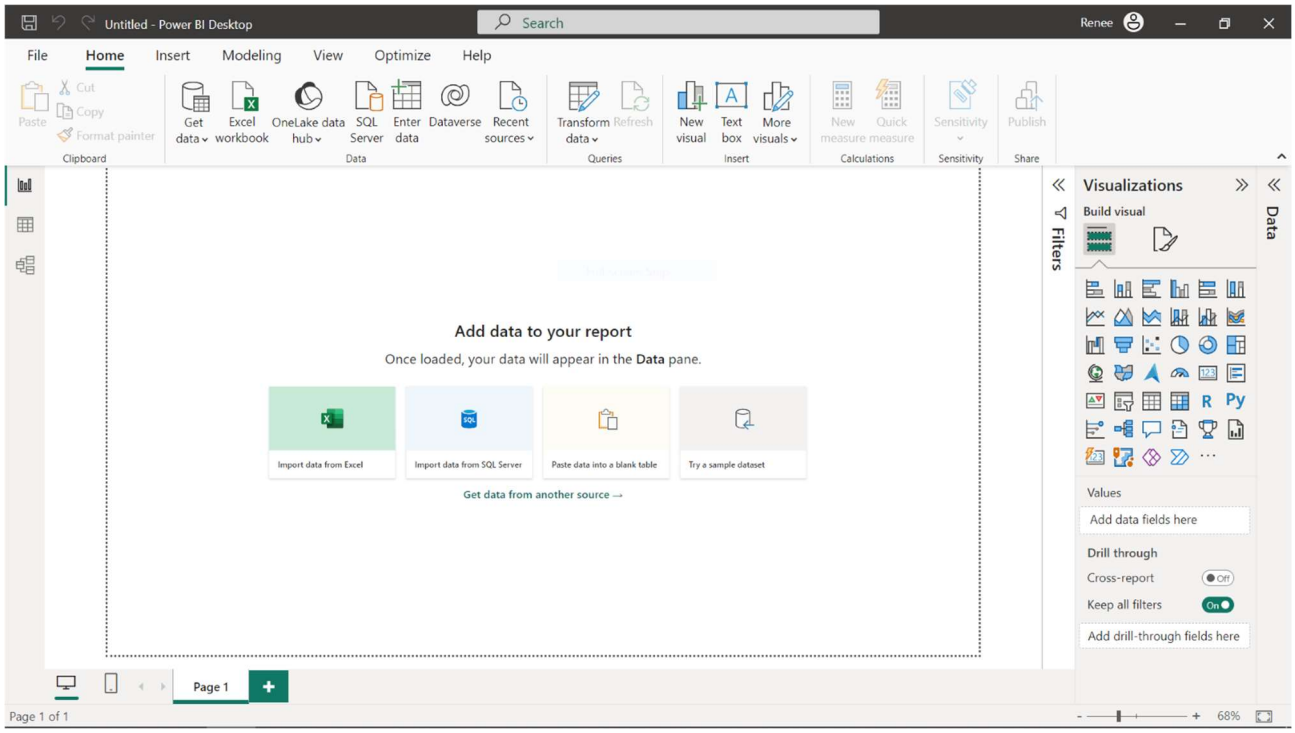
Instructions

Step 1: Connect to the data source

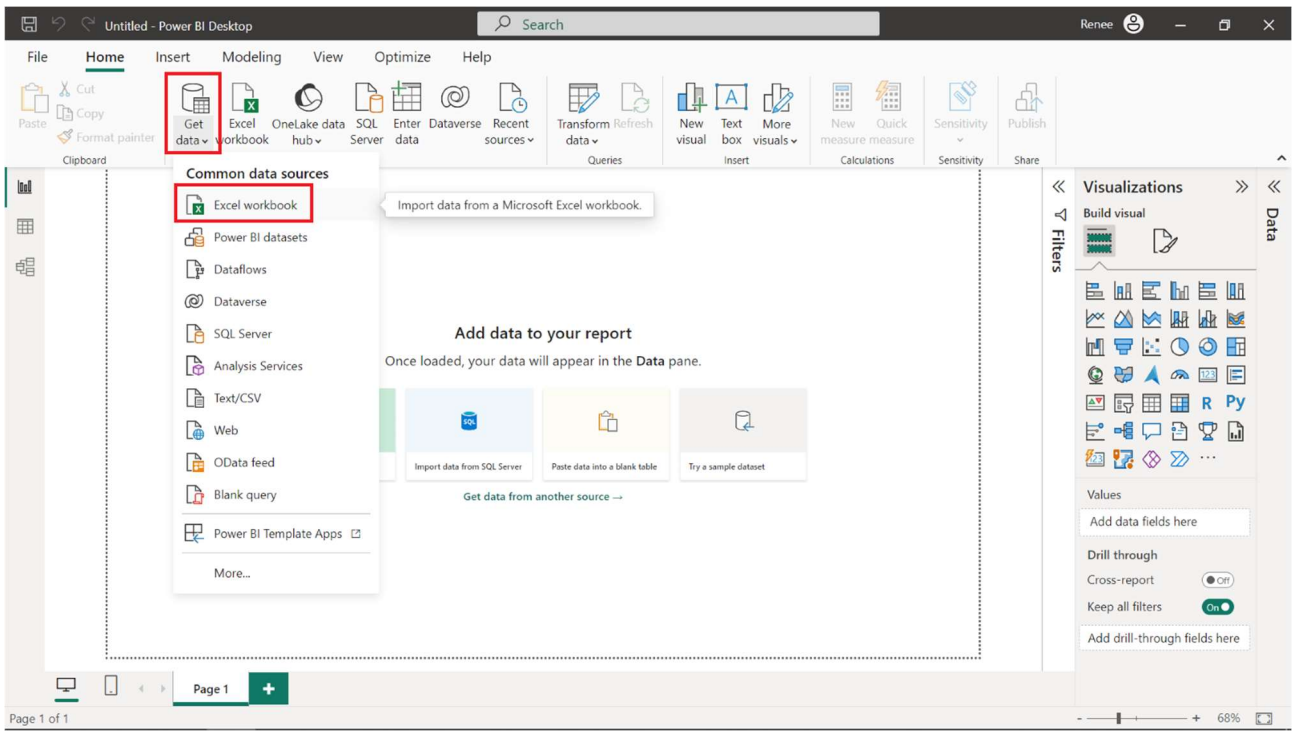
1. Download the *AdventureWorksOrders.xlsx* dataset.

   \module-1_preparing-data\02_profiling-and-cleaning-data\exercises\03_ResolveDataImportErrorsAdventureWorksOrders\datasets\Sj11qWwzQP-rrsZnIBB_tQ_8526cb10b9714809a627d9fbeaf5a6e1_AdventureWorksOrders.xlsx

2. Open Power BI and create a new project.

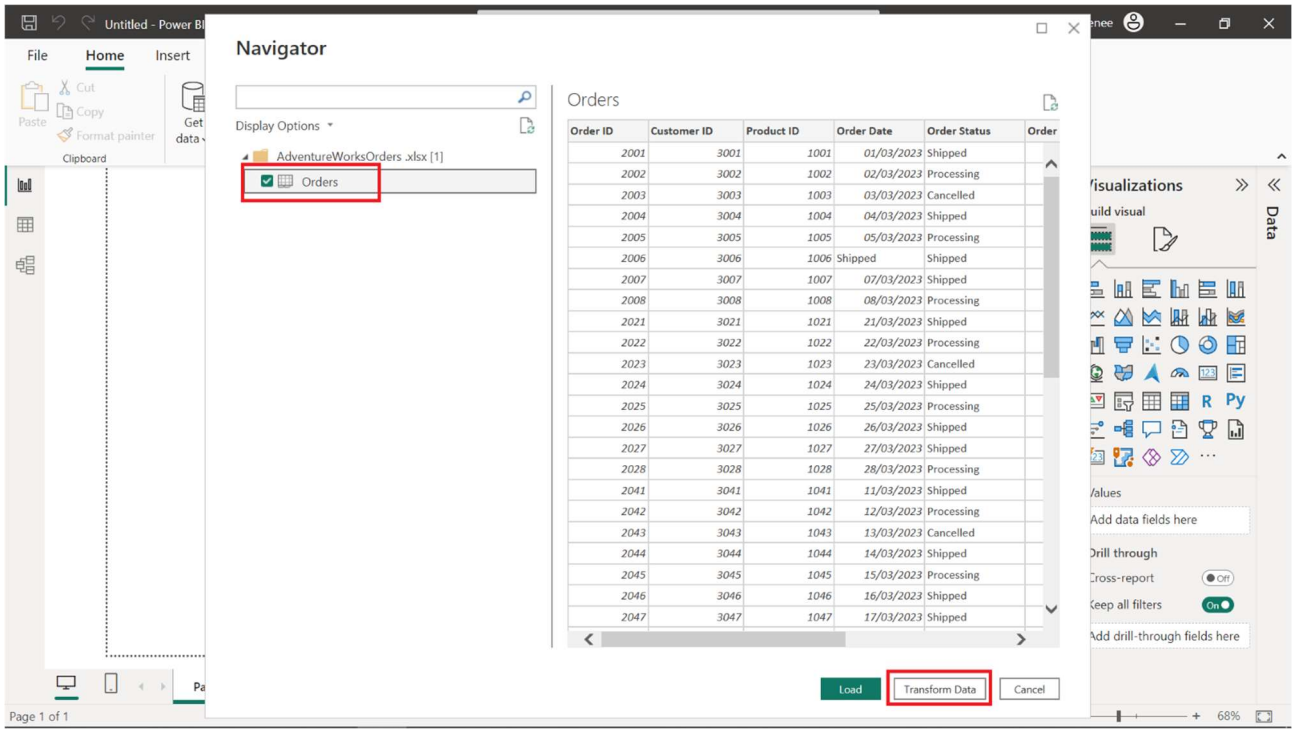


3. On the **Home** tab of the ribbon, select **Get Data** and choose **Excel workbook**.



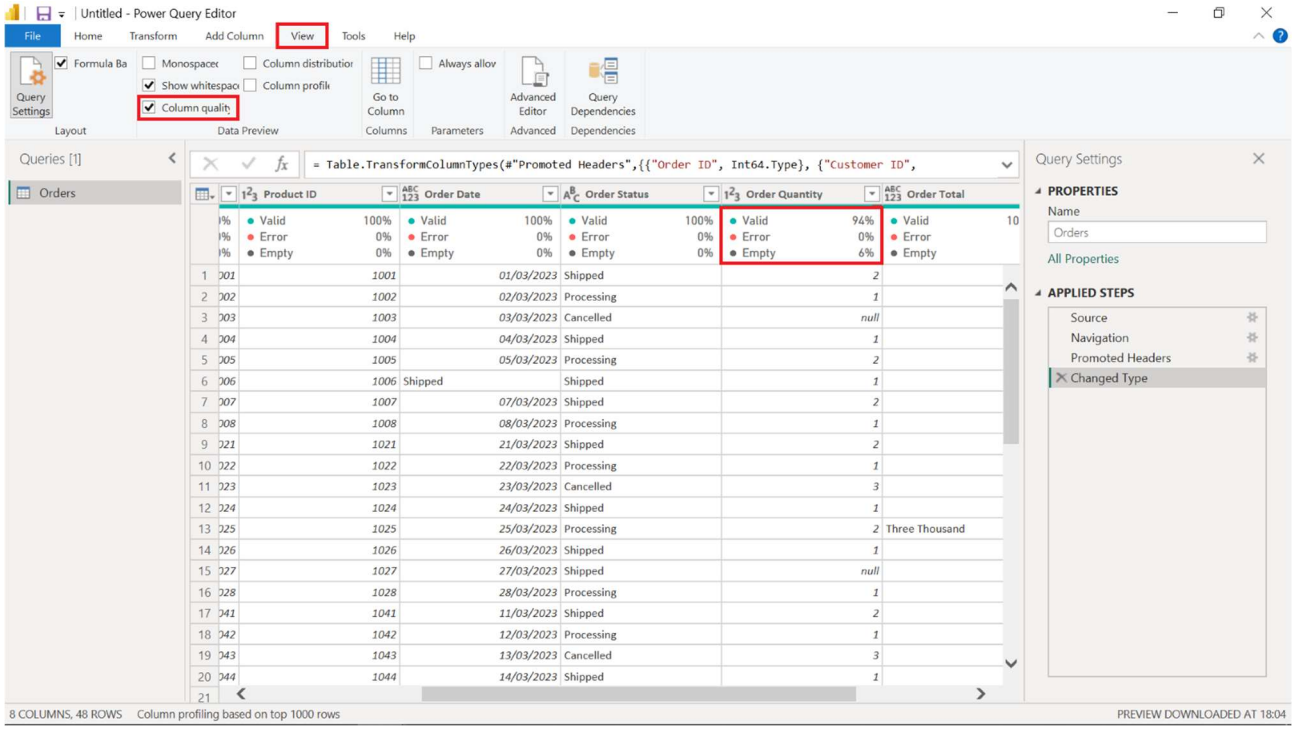
4. When the file browser opens, select the *AdventureWorksOrders.xlsx* file.

5. In the **Navigator** window, select the *AdventureWorksOrders* sheet, followed by **Transform Data.**

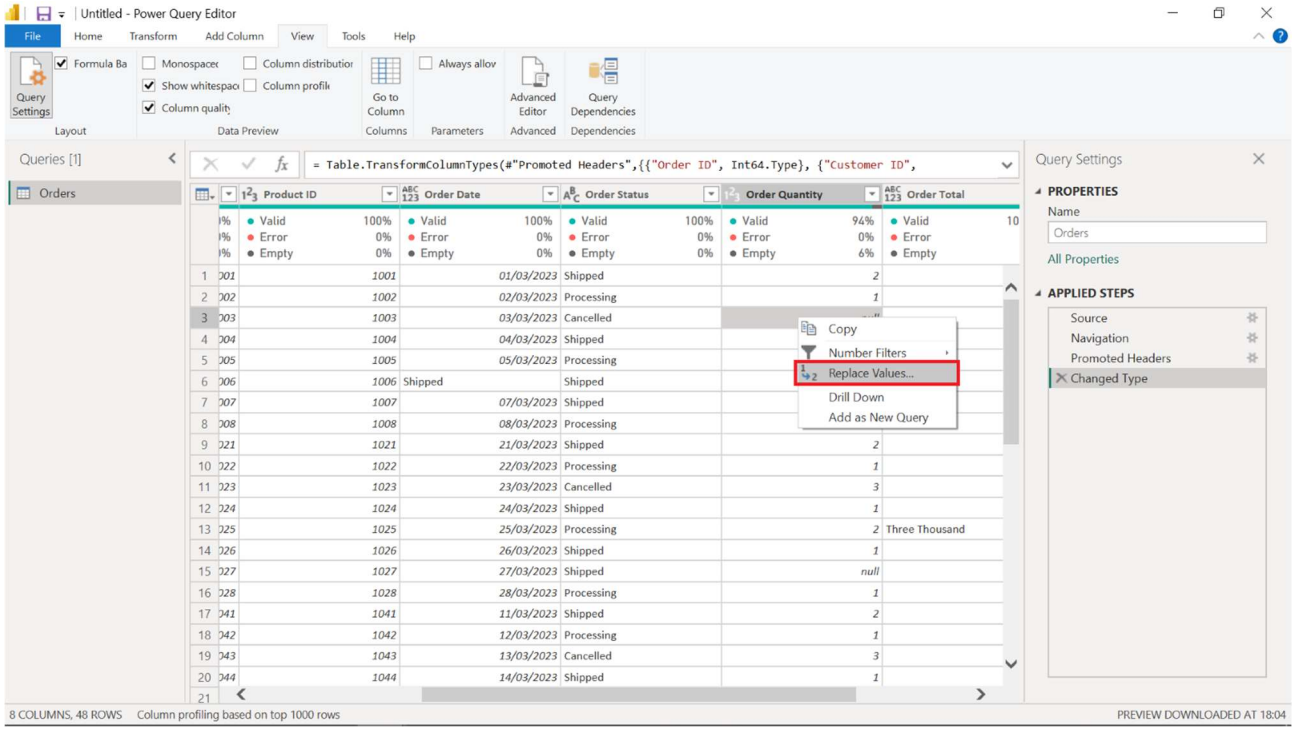


Step 2: Remove null values

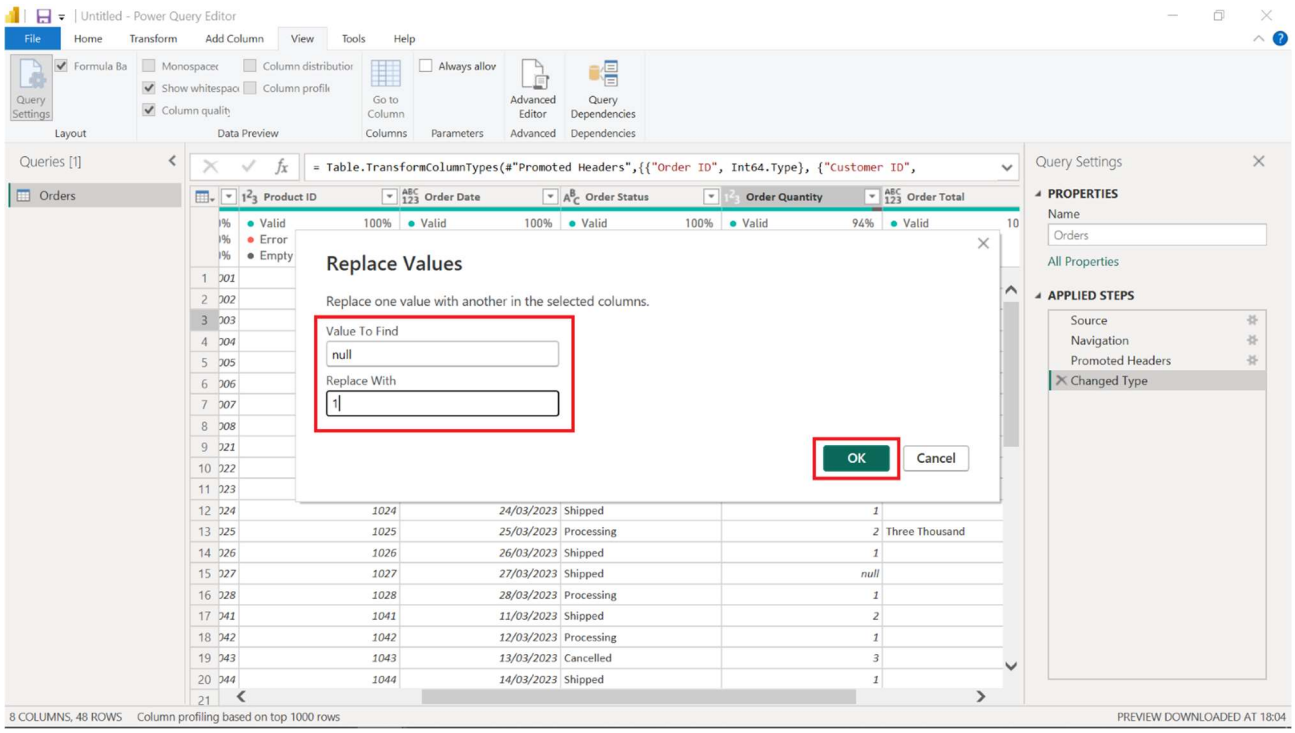
1. When the Power Query Editor opens, go to the **View** tab in the ribbon menu. Enable **Column quality** and inspect the quality values for the **Order Quantity** column. Note that 6% of the values are empty, i.e., null.



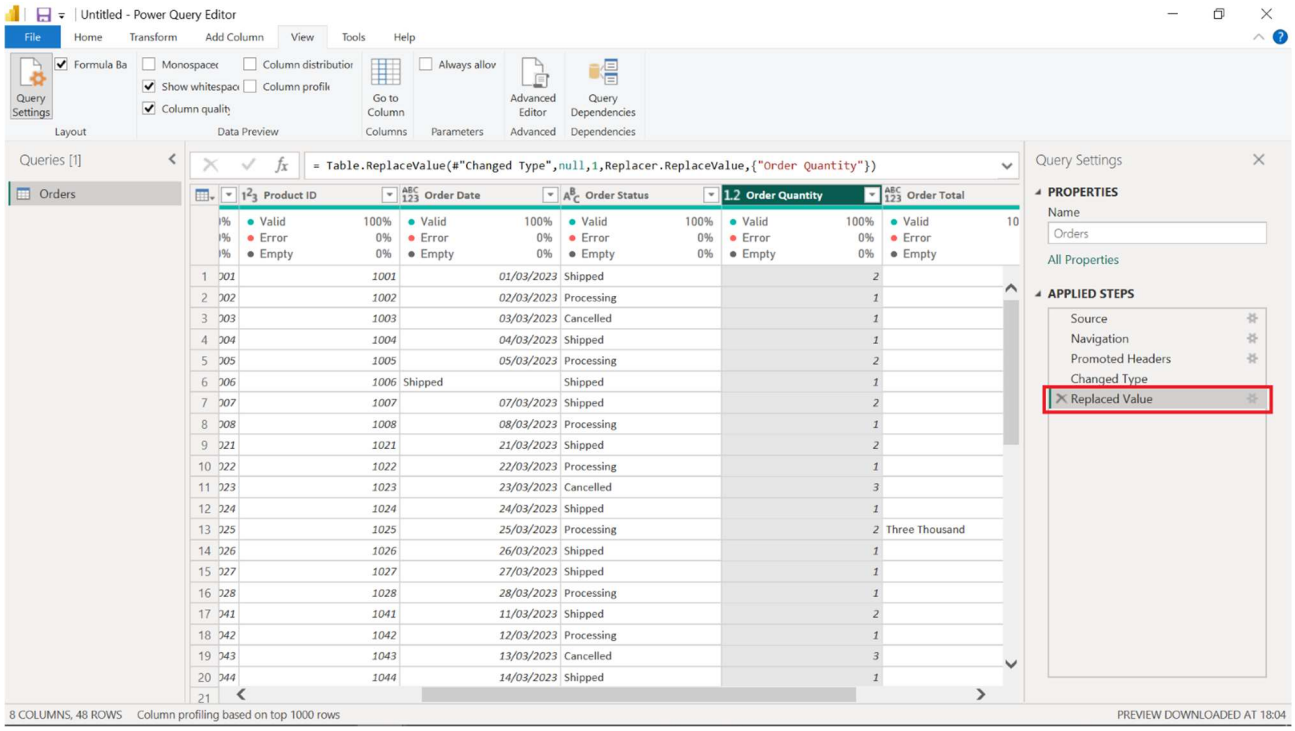
2. The fulfillment team has informed you that the empty values were caused by human error and should be replaced with a value of 1. Right-click one of the null values in the column or the column itself, and select **Replace Values**.



3. In the **Replace Values** prompt, enter **null** as the **Value To Find** and **1** as the value to **Replace With**. Select **OK** and the Power Query Editor updates with the null values replaced.

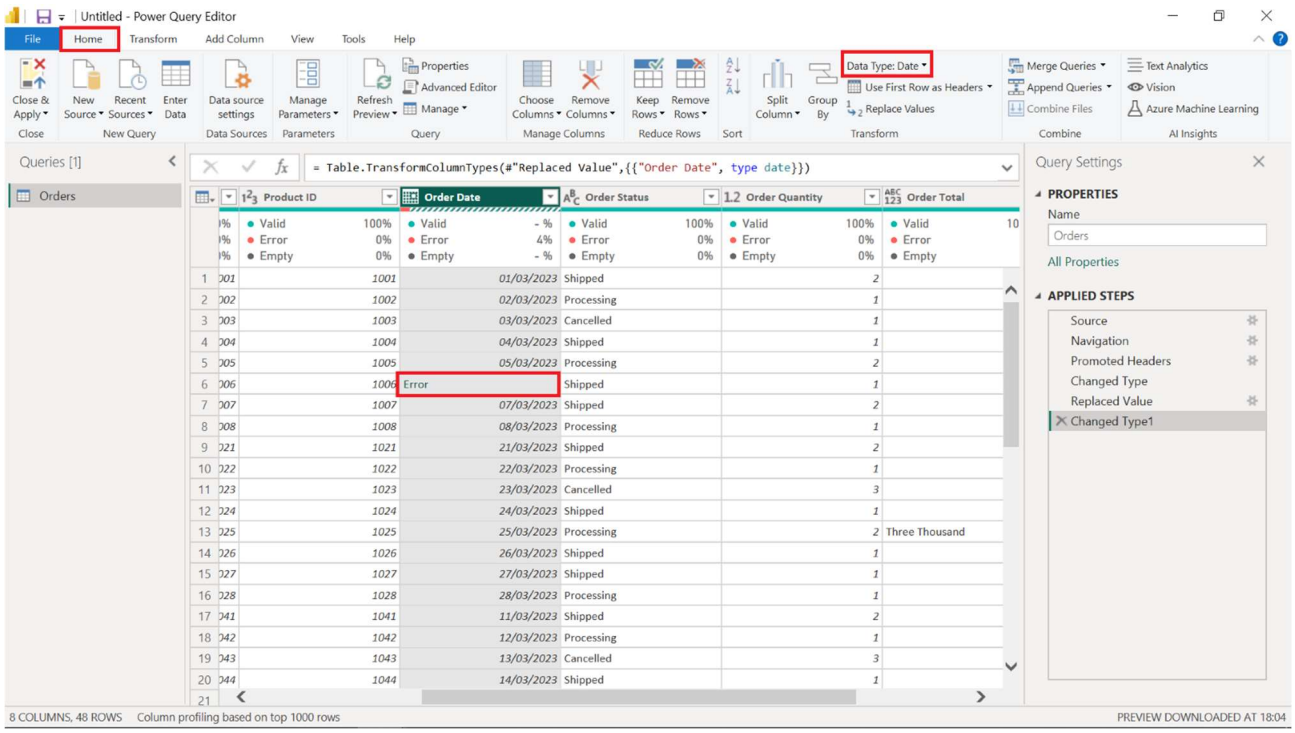


4. Note that a new **Replaced Value** step has been added to the **Applied Steps** list.

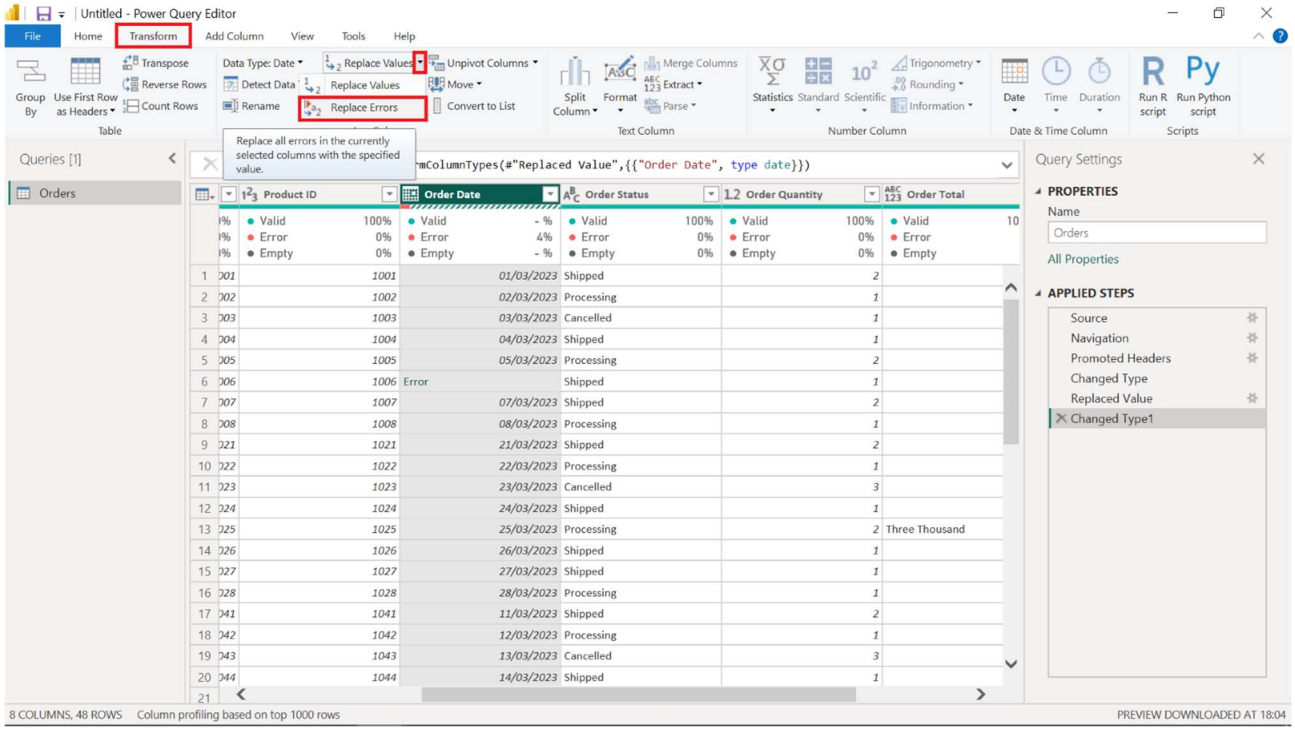


Step 3: Fix data errors in the Order Date column

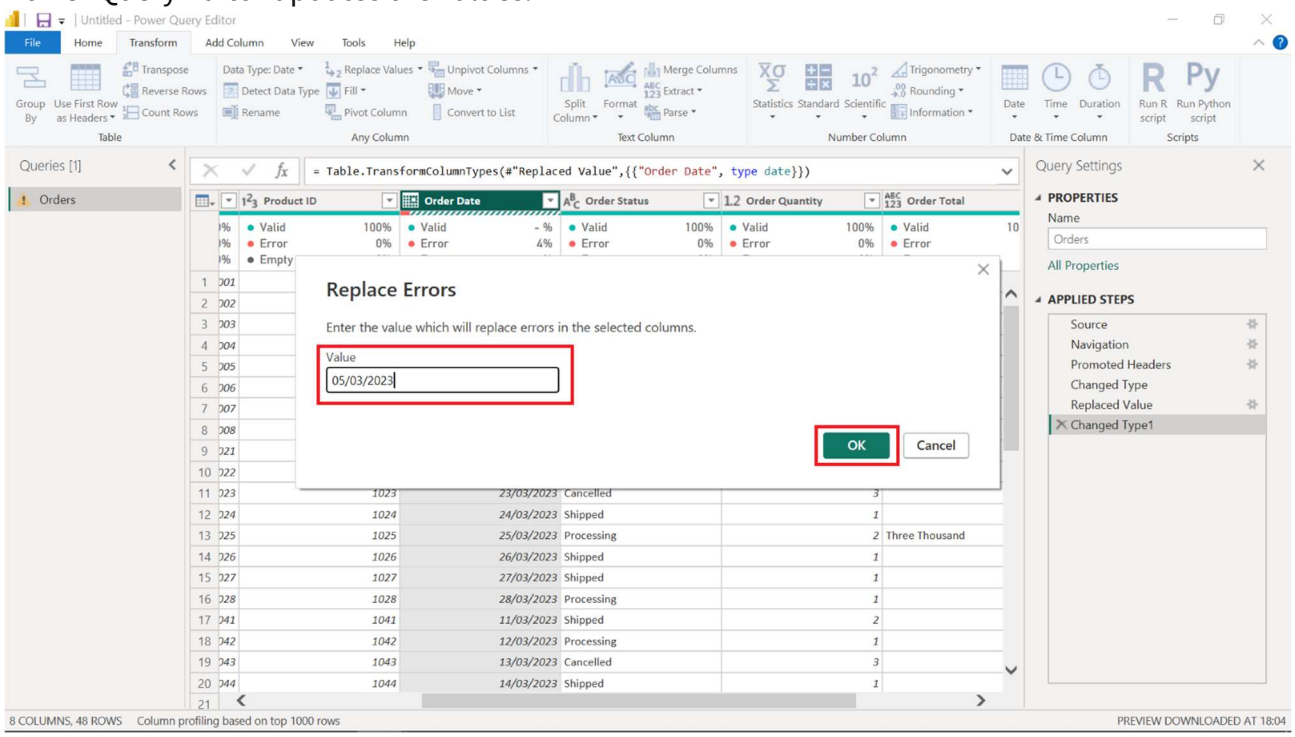
1. Select the **Order Date** column.
2. Navigate to the **Home** tab in the ribbon menu and check the **Data Type** for the column. It is currently set to **Any**, which is incorrect. Change the data type to **Date**. This will add a new **Change Type1** step to the **Applied Steps** list. Note that the column now shows error values. This is due to human error for some values in this column.



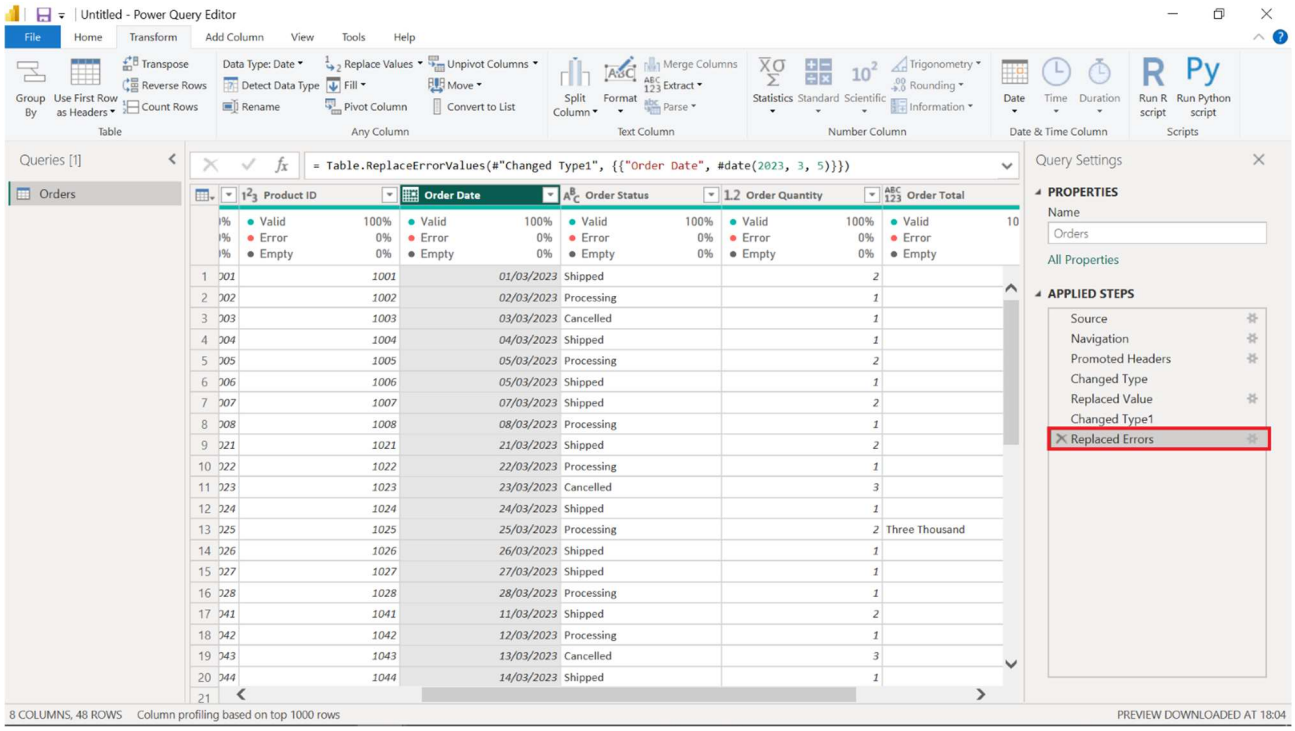
3. The fulfillment team has informed you that these error values should be **05/03/2023**. In the ribbon menu, navigate to the **Transform** tab.
4. Select the dropdown arrow beside **Replace Values** and select **Replace Errors**.



5. In the **Replace Errors** prompt, enter **05/03/2023** as the **Value**. Select **OK**, and the Power Query Editor updates the values.

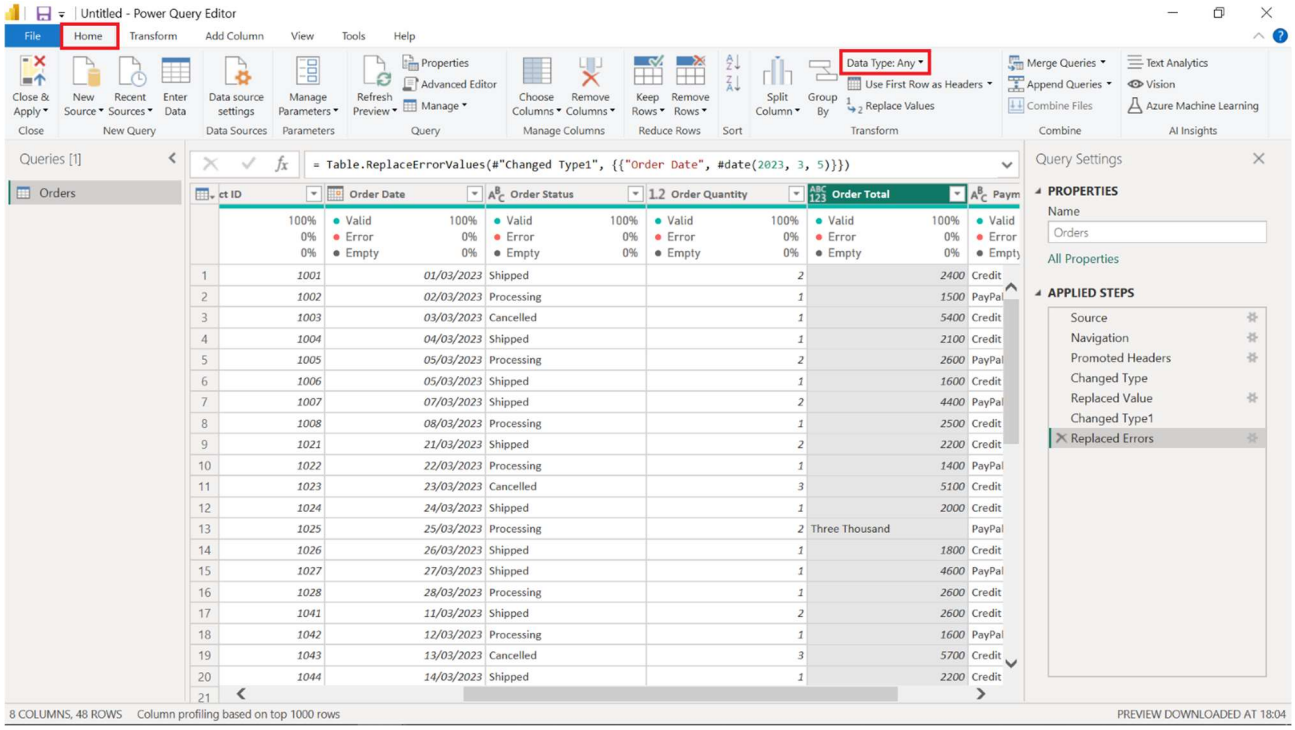


6. Note that a new **Replaced Errors** step has been added to the **Applied Steps** list.

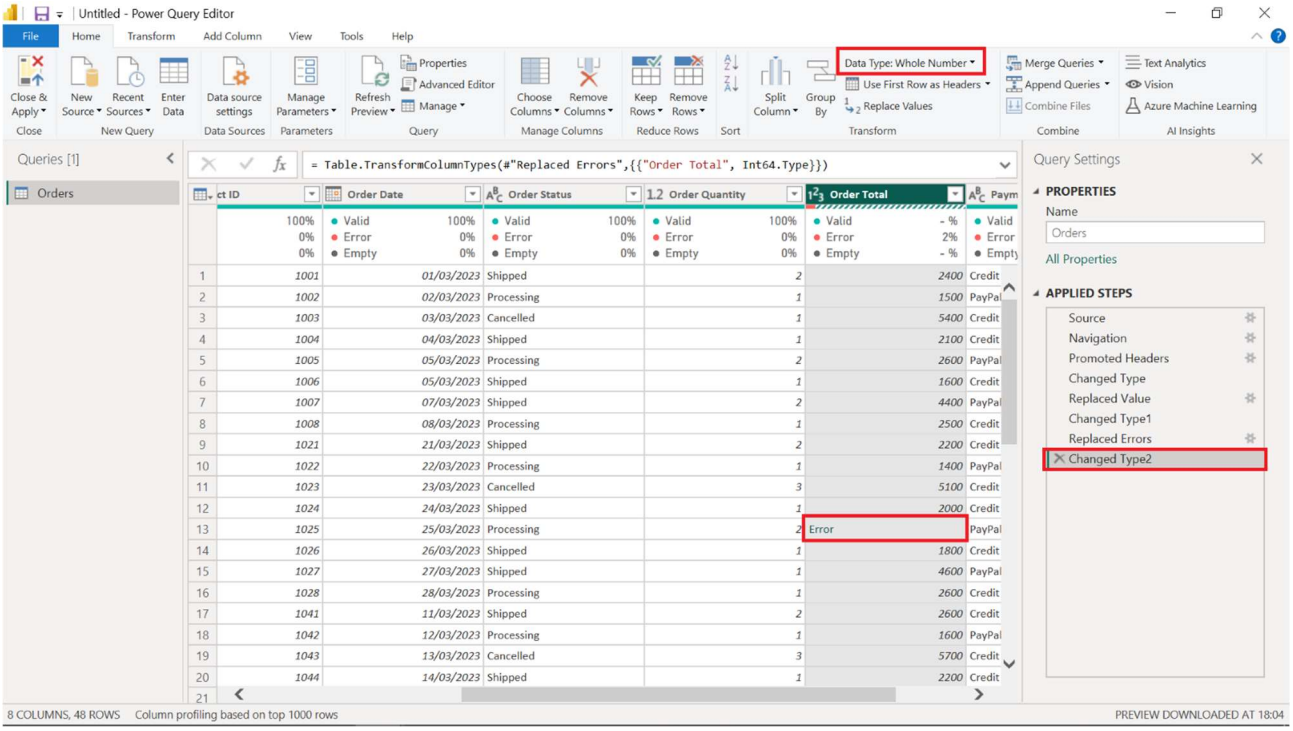


Step 4: Fix data errors in the Order Total column

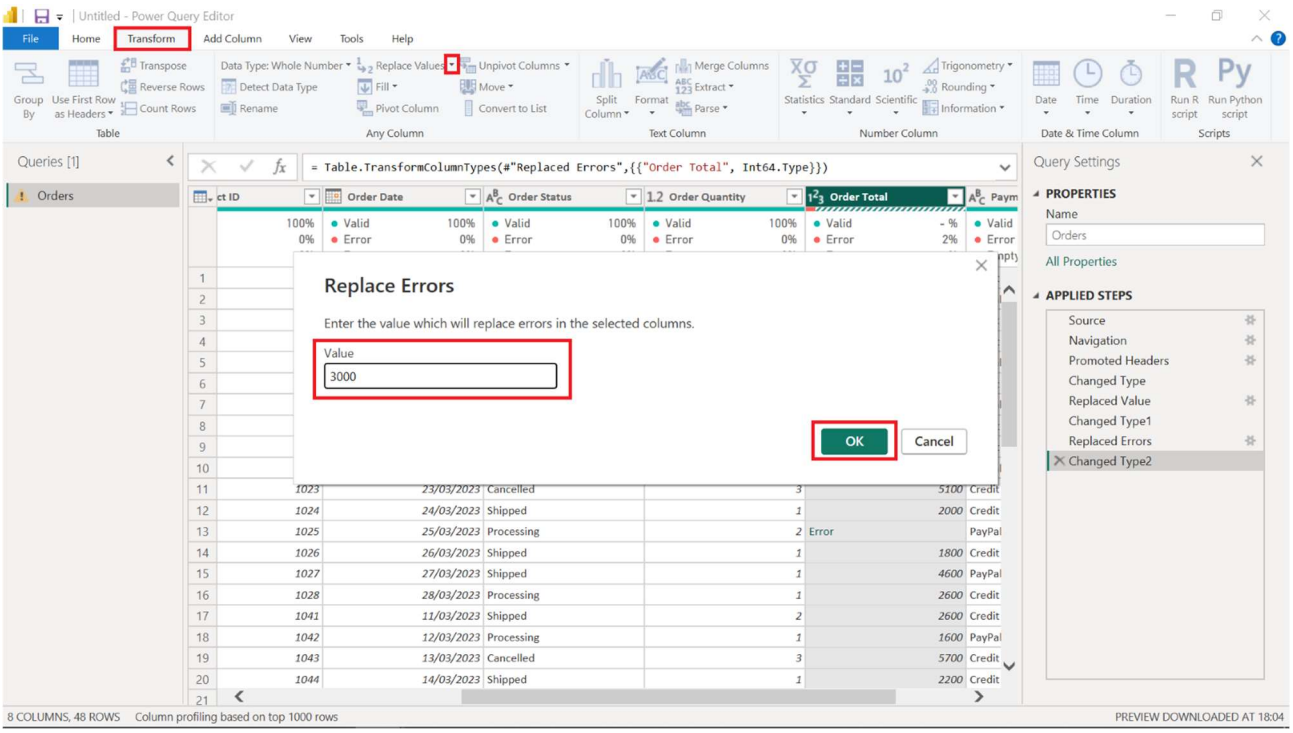
1. Select the **Order Total** column. Navigate to the **Home** tab in the ribbon menu and check the **Data Type** for the column. Again, the column data type is set to **Any**, which is incorrect.



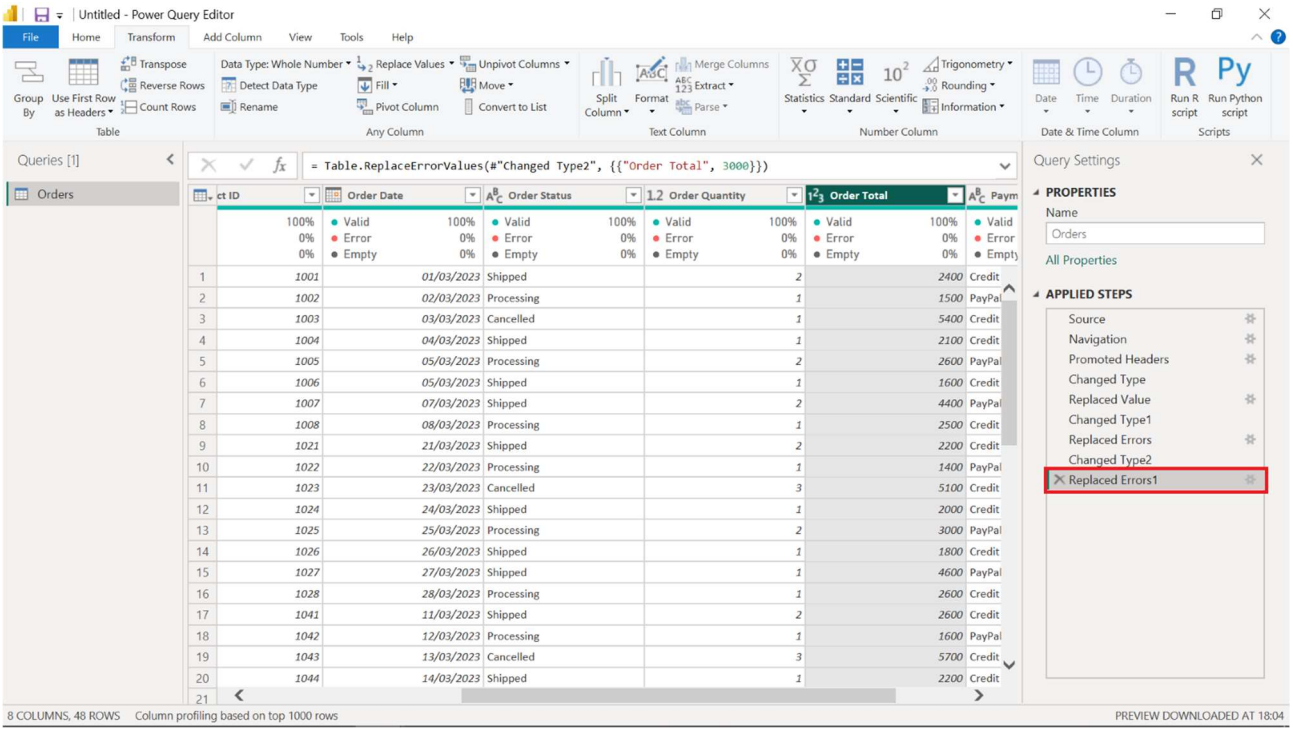
2. Change the data type to **Whole Number**. This will add a new step, **Change Type2,** to the **Applied Steps** list. Again, the column shows error values. This is due to one of the order totals being entered as **Three Thousand** instead of the number 3000.



3. In the ribbon menu, navigate to the **Transform** tab. Again, select the dropdown arrow beside **Replace Values** and select **Replace Errors**. In the **Replace Errors** prompt, enter **3000** as the **Value**.

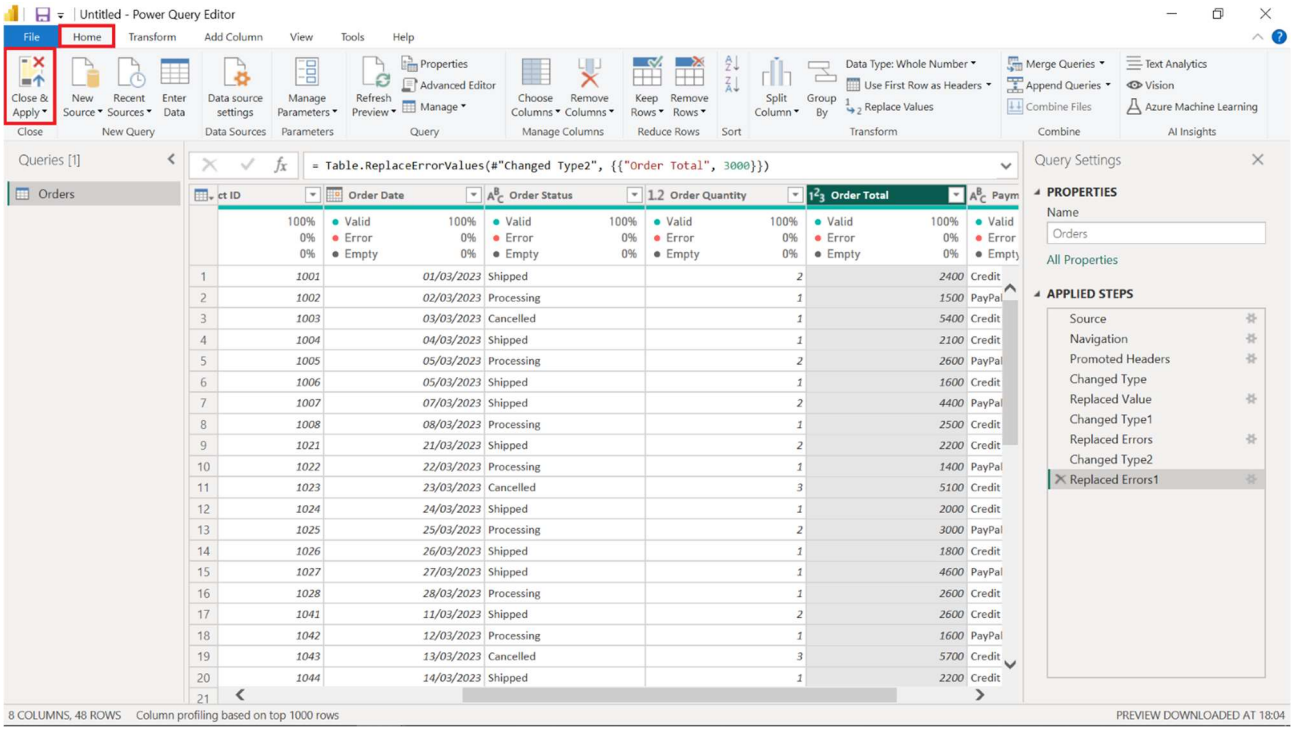


4. Select **OK**, and the Power Query Editor will update. Note that a new **Replaced Errors1** step has been added to the **Applied Steps** list.



Step 5: Import the data

1. Navigate to the **Home** tab in the ribbon menu and select **Close & Apply**. Power BI will process the data and successfully complete the import.



2. Save your project and close Power BI.

Conclusion

Congratulations! You have successfully resolved data import errors using the Power Query Editor. The data is now ready to be analyzed in Power BI. Working with error-free data is important to a successful data analysis process.